



Lecture 3: EM Wave Polarization

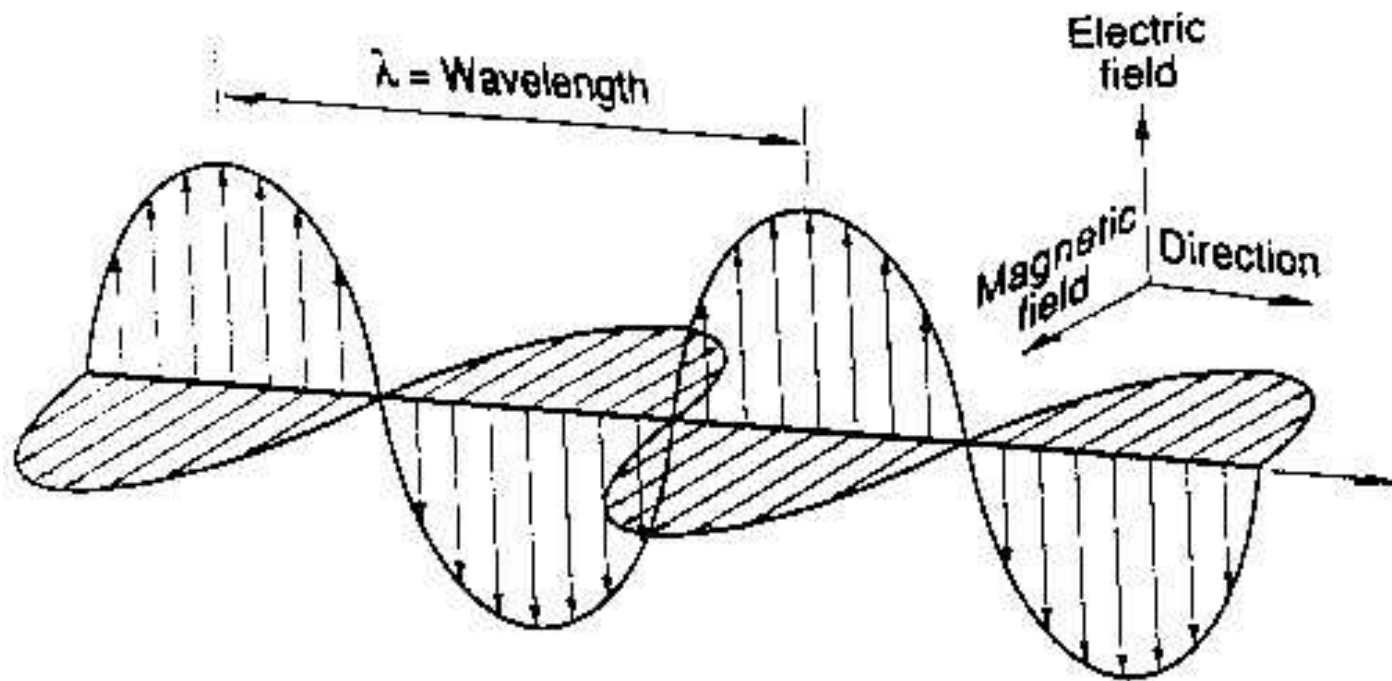
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Linearly polarized Wave

A plane EM wave is transverse. For a linearly polarized wave, the **electric vector** simply oscillates in a certain direction, which, with the **propagation direction**, defines **the plane of polarization**.



Vector Wave

For a plane EM wave propagating in the z direction, the x and y components of both $\mathbf{E}$ and $\mathbf{H}$ obey the same wave eq. So, it suffices to discuss only $\mathbf{E}$. In general, the x and y components are different in amplitude with different phases, i.e., **2 orthogonal linearly polarized components**,

$$E_x = A_x \cos(kz - \omega t + \delta_x)$$

$$E_y = A_y \cos(kz - \omega t + \delta_y)$$

$$E_z = 0$$

here $k = 2\pi/\lambda$, where λ is the wavelength in cm, and $\omega = 2\pi\nu$, where ν is freq. in Hz. E_x and E_y are related and can be packed together with the phase defined as

$$\tau = kz - \omega t$$

Rewriting the eqs.,

$$\frac{E_x}{A_x} = \cos \tau \cos \delta_x - \sin \tau \sin \delta_x$$

$$\frac{E_y}{A_y} = \cos \tau \cos \delta_y - \sin \tau \sin \delta_y$$

Vector waves - Elliptical Polarization

From previous slide,

$$\frac{E_x}{A_x} = \cos \tau \cos \delta_x - \sin \tau \sin \delta_x$$
$$\frac{E_y}{A_y} = \cos \tau \cos \delta_y - \sin \tau \sin \delta_y$$

Multiplying with $\sin \delta_x$, $\sin \delta_y$, $\cos \delta_x$, $\cos \delta_y$, and then subtracting, we have

$$\frac{E_x}{A_x} \sin \delta_y - \frac{E_y}{A_y} \sin \delta_x = \cos \tau \sin(\delta_y - \delta_x)$$
$$\frac{E_x}{A_x} \cos \delta_y - \frac{E_y}{A_y} \cos \delta_x = \sin \tau \sin(\delta_y - \delta_x)$$

Squaring and adding we obtain

$$\left(\frac{E_x}{A_x}\right)^2 + \left(\frac{E_y}{A_y}\right)^2 - 2 \frac{E_x}{A_x} \frac{E_y}{A_y} \cos \delta = \sin^2 \delta$$

with

$$\delta = \delta_x - \delta_y$$

Vector waves - Elliptical Polarization

From previous slide,

$$\left(\frac{E_x}{A_x}\right)^2 + \left(\frac{E_y}{A_y}\right)^2 - 2\frac{E_x}{A_x}\frac{E_y}{A_y}\cos\delta = \sin^2\delta$$

with

$$\delta = \delta_x - \delta_y$$

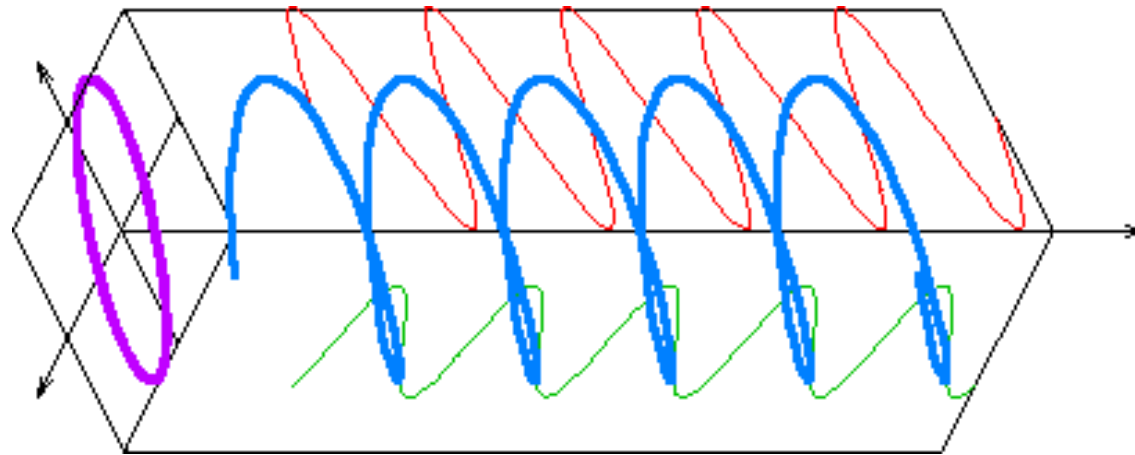
$\Rightarrow$ an ellipse and the wave is **elliptically polarized**, for both the electric and magnetic field of the wave; $\sin\delta$ determines the sense in the the electric vector rotates.

$\sin\delta > 0$, the x component starts earlier than the y component, thus **$\mathbf{E}$ vector rotates** from +x-axis to +y-axis $\Rightarrow$ **right-handed** for which the rotation of the **$\mathbf{E}$ vector** and the direction of propagation form a right-handed screw.

$\sin\delta < 0$ for **left-handed** elliptical polarization.

[Rybicki (Opposite sense): Right-handed if the electric field is rotating clockwise as seen by an observer towards whom the wave is moving.]

Left-Handed Elliptically Polarized Wave



In electrodynamics, elliptical polarization is the polarization of electromagnetic radiation such that the tip of the electric field vector describes an ellipse in any fixed plane intersecting, and normal to, the direction of propagation. An elliptically polarized wave may be resolved into two linearly polarized waves in phase quadrature with their polarization planes at right angles (i.e., orthogonal) to each other. This figure shows a [left-handed elliptically polarized wave](#).

Two Special cases

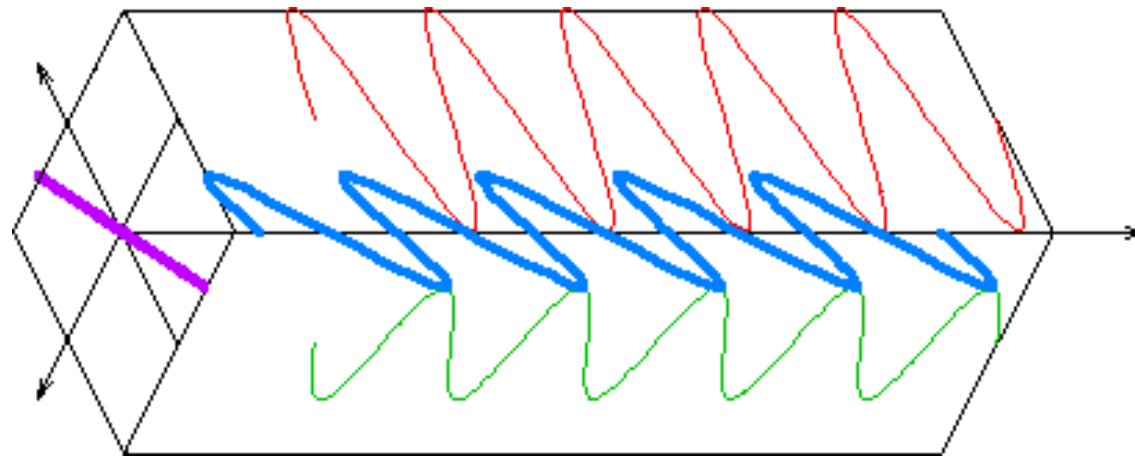
(1) If $\delta = \delta_x - \delta_y = m\pi$, $m = 0, \pm 1, \pm 2..$

Then

$$E_x = A_x \cos(\tau + \delta_x)$$

$$E_y = \pm A_y \cos(\tau + \delta_x)$$

The polarization ellipse degenerates into a straight line and $\mathbf{E}$ is linearly polarized.



Demo

http://webphysics.davidson.edu/physlet_resources/dav_optics/Examples/polarization.html

Two Special cases

(2) If

$$A_x = A_y = A$$

and

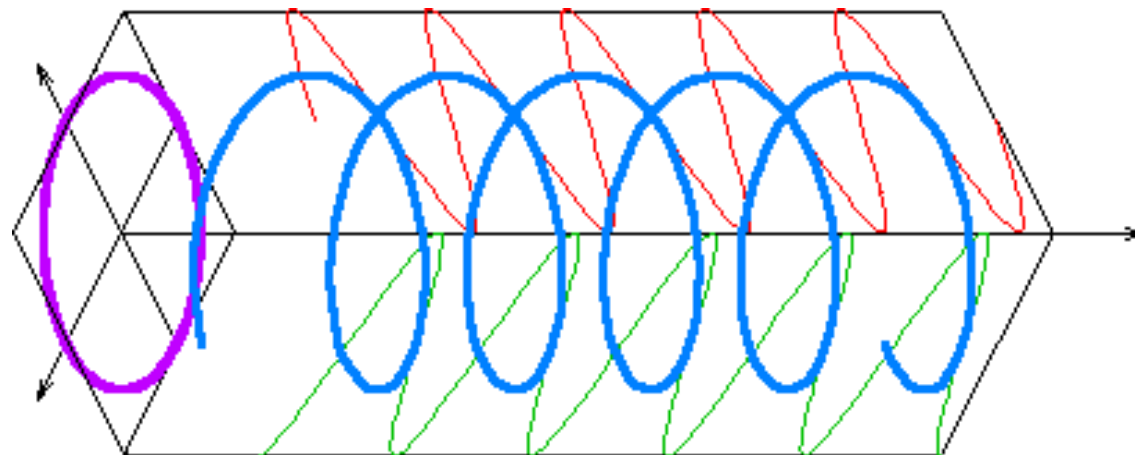
$$\delta = \frac{\pi}{2}(1 + m), \quad m = 0, \pm 1, \pm 2..$$

⇒ circularly polarized wave. In this case,

$$E_x = A \cos(\tau + \delta_x)$$

$$E_y = \pm A \sin(\tau + \delta_x)$$

$$E^2 = E_x^2 + E_y^2 = A^2 = \text{constant}$$



Choosing the right coor. system

The geometric properties of the ellipse are seen best by selecting a coordinate system oriented along the major and minor axes, i.e., decomposing the elliptically polarized wave into **two orthogonal linearly polarized waves along the major and minor axes (ξ, η)**. In this system, the two components have a phase difference of $\pi/2$ and can be assumed to be given by the following

$$E_{\xi} = A_{\xi} \cos(\tau + \delta)$$

$$E_{\eta} = A_{\eta} \sin(\tau + \delta)$$

With ψ being the angle betw. the two coordinate systems (x, y) and (ξ, η) ,

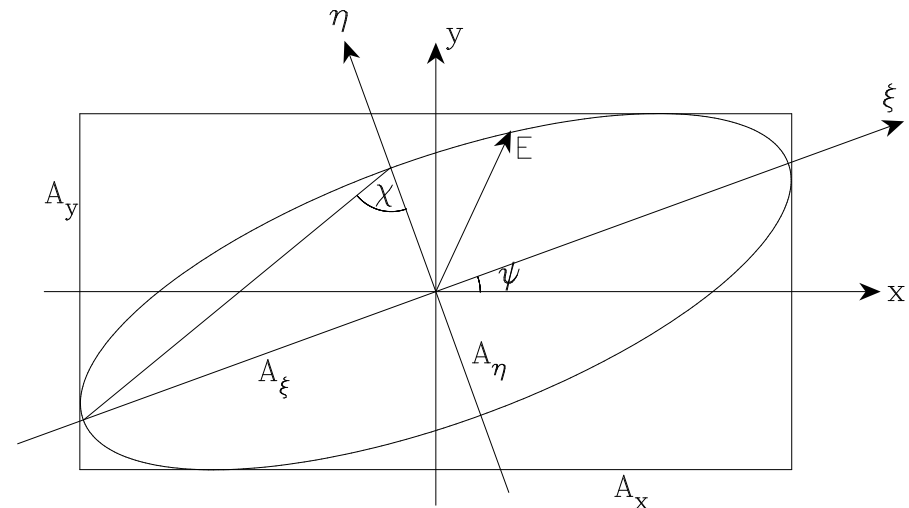
$$E_{\xi} = E_x \cos \psi + E_y \sin \psi$$

$$E_{\eta} = -E_x \sin \psi + E_y \cos \psi$$

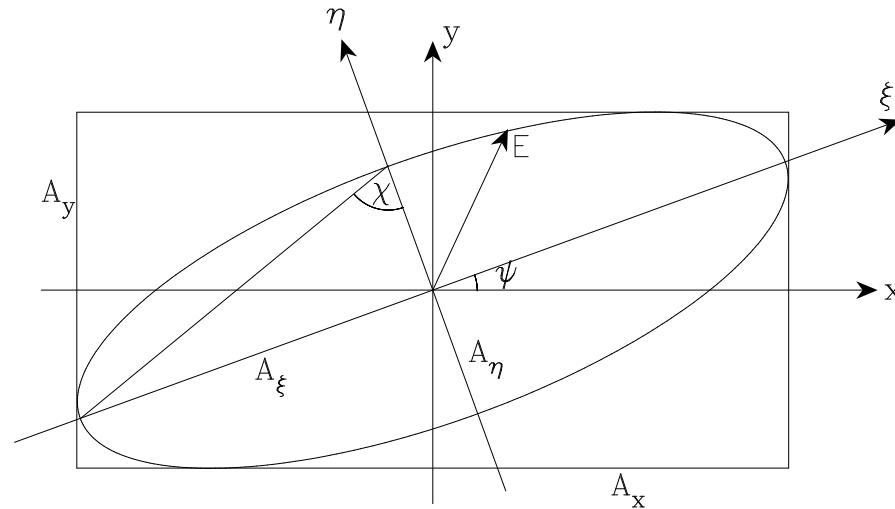
Thus,

$$E_{\xi} = A_{\xi} \cos(\tau + \delta) = E_x \cos \psi + E_y \sin \psi$$

$$E_{\eta} = A_{\eta} \sin(\tau + \delta) = -E_x \sin \psi + E_y \cos \psi$$



Two orthogonal components - cont



The values in the two coordinate systems have the following relationship:

$$S_0 \equiv A_\xi^2 + A_\eta^2 = A_x^2 + A_y^2$$

and

$$\tan 2\psi = -\tan 2\alpha \cos \delta, \quad \sin 2\chi = \sin 2\alpha \sin \delta$$

with

$$\frac{A_x}{A_y} = \tan \alpha, \quad \frac{A_\xi}{A_\eta} = \tan \chi$$

The total poynting flux of the polarized wave $\propto S_0$, and is equal to the sum of the fluxes of two orthogonal, but otherwise arbitrary directions. $A_x, A_y, \delta \Leftrightarrow A_\xi, A_\eta, \psi$.

Two Circularly polarized waves

Indeed, an arbitrary elliptically polarized wave can be decomposed into the sum of **two circularly polarized waves** at any time.

$$\begin{aligned}E_{\xi} &= A_{\xi} \cos(\tau + \delta) = (A_r + A_l) \cos(\tau + \delta) \\E_{\eta} &= A_{\eta} \sin(\tau + \delta) = (A_r - A_l) \cos(\tau + \delta - \frac{\pi}{2})\end{aligned}$$

with the two circularly polarized waves

$$\begin{aligned}\mathbf{E}_r &= A_r [\cos(\tau + \delta) \hat{\xi} + \cos(\tau + \delta - \frac{\pi}{2}) \hat{\eta}] \\ \mathbf{E}_l &= A_l [\cos(\tau + \delta) \hat{\xi} - \cos(\tau + \delta - \frac{\pi}{2}) \hat{\eta}]\end{aligned}$$

$$A_r = \frac{1}{2}(A_{\xi} + A_{\eta}), \quad A_l = \frac{1}{2}(A_{\xi} - A_{\eta})$$

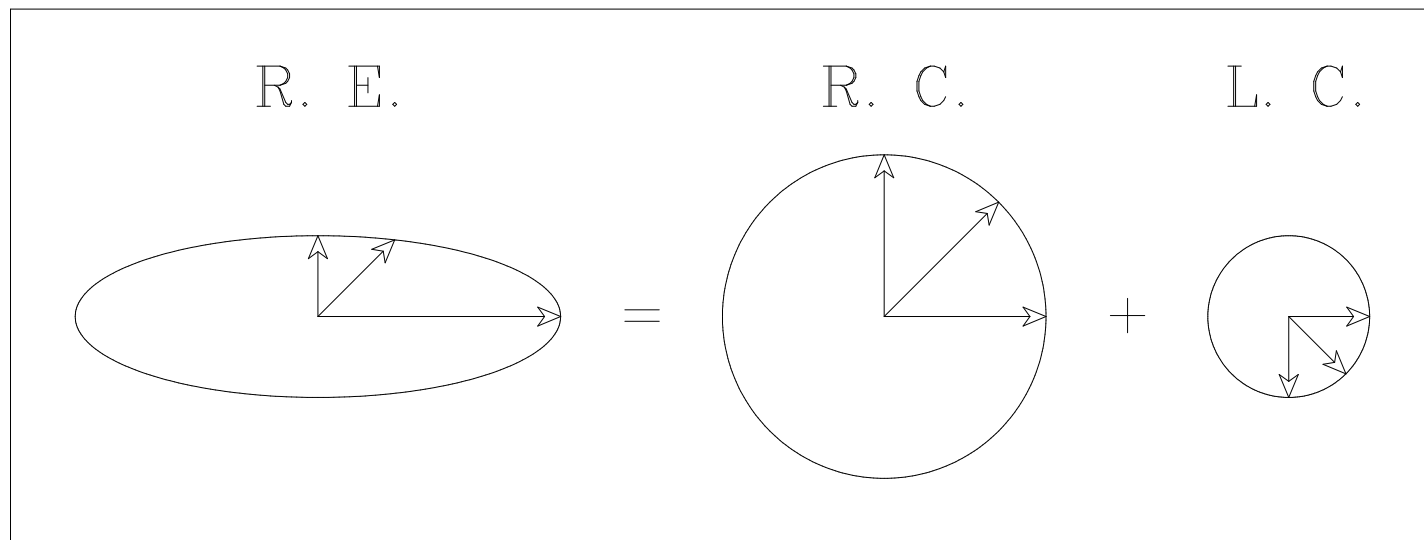
The total Poynting flux of the wave

$$S_0 = A_{\xi}^2 + A_{\eta}^2 = A_r^2 + A_l^2$$

Thus, $A_x, A_y, \delta \Leftrightarrow A_{\xi}, A_{\eta}, \psi \Leftrightarrow A_r, A_l, \delta$.

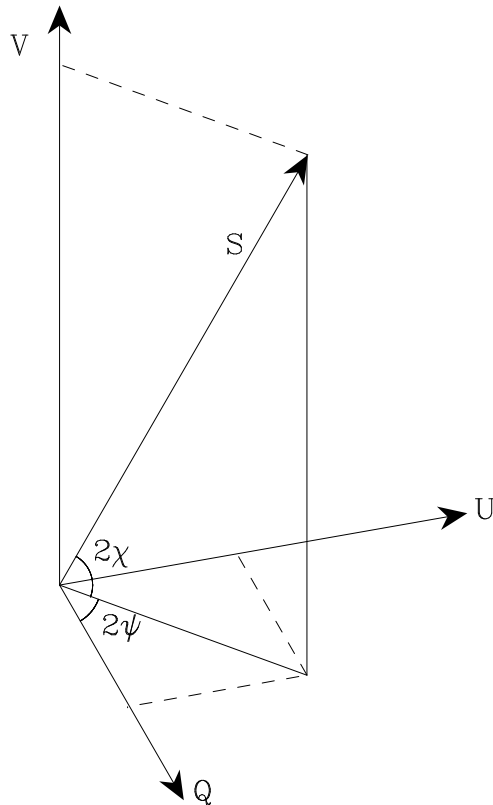
Two Circularly polarized waves

E.g. a right-handed elliptically polarized wave can be decomposed as follows:



Poincaré Sphere and Stokes parameters

Poincaré (1892) introduced an easy visualization of all the different states of polarization of a vector wave by interpreting the angles 2ψ as longitude and 2χ as latitude on a sphere with the radius S_0 . The Stokes parameters (I, Q, U, V) are:



$$S_0 = I = A_\xi^2 + A_\eta^2$$

$$S_1 = Q = S_0 \cos 2\chi \cos 2\psi$$

$$S_2 = U = S_0 \cos 2\chi \sin 2\psi$$

$$S_3 = V = S_0 \sin 2\chi$$

with

$$\sin 2\chi = V/I$$

$$\tan 2\psi = U/Q$$

$$S_0^2 = S_1^2 + S_2^2 + S_3^2$$

$$I^2 = Q^2 + U^2 + V^2$$

Here, I or S_0 is nonnegative and is proportional to the total energy flux or intensity of the wave.

Also, Q, U, V forms a right-handed screw.

Stokes Parameters

The Stokes parameters can be converted into observable quantities:

$$S_0 = I = A_x^2 + A_y^2$$

$$S_1 = Q = A_x^2 - A_y^2$$

$$S_2 = U = 2A_x A_y \cos \delta$$

$$S_3 = V = 2A_x A_y \sin \delta$$

All waves are ellipt. polarized with 3 special cases:

(1) Right-handed circularly polarized

$$A_x = A_y, \delta = \pi/2 \Rightarrow I = S, Q = U = 0, V = S$$

(2) Left-handed circularly polarized

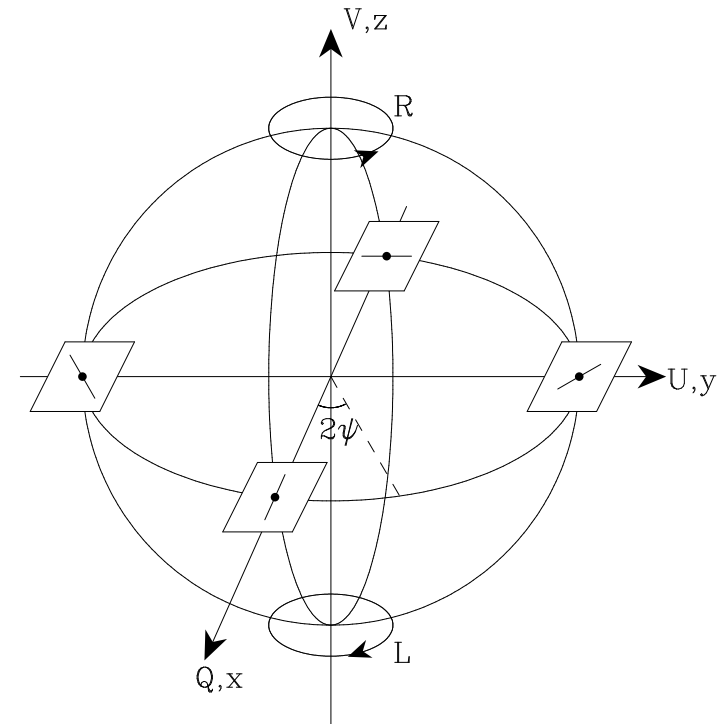
$$A_x = A_y, \delta = -\pi/2 \Rightarrow I = S, Q = U = 0, V = -S$$

(3) Linearly polarized

$$A_\eta = E, A_\xi = 0, \chi = 0$$

$$\Rightarrow I = S, Q = I \cos 2\psi, U = I \sin 2\psi, V = 0$$

Above figure assumes (Q, U, V) aligned with (x, y, z) and the wave propagates in z direction. Note the factors of two before ψ and χ corresponding respectively to the facts that any polarization ellipse is indistinguishable from one rotated by 180° , or one with the semi-axis lengths swapped accompanied by a 90° rotation.



Quasi-Monochromatic Plane waves

A strictly **monochromatic wave** is always (completely or 100%) polarized since the $\mathbf{E}$ vector displays a simple, nonrandom directional behavior in time, i.e, A_x , A_y , δ_x and δ_y are always constant. A EM wave usually has a small but finite bandwidth and is **quasi-monochromatic wave** if $\Delta\nu \ll \bar{\nu}$. Suppose we have

$$E_x(t) = a_x(t)e^{i(\phi_x(t)-2\pi\bar{\nu}t)}, \quad E_y(t) = a_y(t)e^{i(\phi_y(t)-2\pi\bar{\nu}t)}$$

E_x and E_y are the analytic signals associated with the **real components**:

$$E_x^{(r)}(t) = a_x(t) \cos(\phi_x(t) - 2\pi\bar{\nu}t), \quad E_y^{(r)}(t) = a_y(t) \cos(\phi_y(t) - 2\pi\bar{\nu}t)$$

Then, with $\delta = (\phi_x - \phi_y)$, **the time averaged values**:

$$\begin{aligned} S_0 &= I \equiv \langle E_x E_x^* \rangle + \langle E_y E_y^* \rangle = \langle a_x^2 \rangle + \langle a_y^2 \rangle \\ S_1 &= Q \equiv \langle E_x E_x^* \rangle - \langle E_y E_y^* \rangle = \langle a_x^2 \rangle - \langle a_y^2 \rangle \\ S_2 &= U \equiv \langle E_x E_y^* \rangle + \langle E_y E_x^* \rangle = 2\langle a_x a_y \cos \delta \rangle \\ S_3 &= V \equiv (\langle E_x E_y^* \rangle - \langle E_y E_x^* \rangle)/i = 2\langle a_x a_y \sin \delta \rangle \end{aligned}$$

Stokes parameters - cont.

According to the Schwartz inequality, we have

$$\langle E_x E_x^* \rangle \langle E_y E_y^* \rangle \geq \langle E_x E_y^* \rangle \langle E_y E_x^* \rangle$$

the equality sign holds only when the ratio of $E_x(t)$ to $E_y(t)$ is a complex constant, independent of time, e.g, for monochromatic wave. Thus, we have

$$\begin{aligned} S_0^2 &\geq S_1^2 + S_2^2 + S_3^2 \\ I^2 &\geq Q^2 + U^2 + V^2 \end{aligned}$$

The equality holds for a completely elliptically polarized wave. For an **unpolarized wave**, where the phases between E_x and E_y maintain no permanent relation and where there is no preferred orientation in the x - y plane, so that $\langle a_x^2 \rangle = \langle a_y^2 \rangle$ and thus

$$Q = U = V = 0$$

Stokes Parameters : Superposition

The Stokes parameters of the superposition of several independent vector waves will be the sum of the Stokes parameters of the individual waves. For a superposition of different waves, each having its own $E_x^{(k)}$ and $E_y^{(k)}$, $k = 1, 2, 3, \dots$, we have

$$E_x = \sum_k E_x^{(k)}, \quad E_y = \sum_k E_y^{(k)}$$

so that

$$\langle E_i E_j^* \rangle = \sum_k \sum_l \langle E_i^{(k)} E_j^{(l)*} \rangle = \sum_k \langle E_i^{(k)} E_j^{(k)*} \rangle$$

here only terms with $k = l$ survive the averaging because of the random phases. Thus,

$$\begin{aligned} I &= \sum I^{(k)} \\ Q &= \sum Q^{(k)} \\ U &= \sum U^{(k)} \\ V &= \sum V^{(k)} \end{aligned}$$

Degree of Polarization

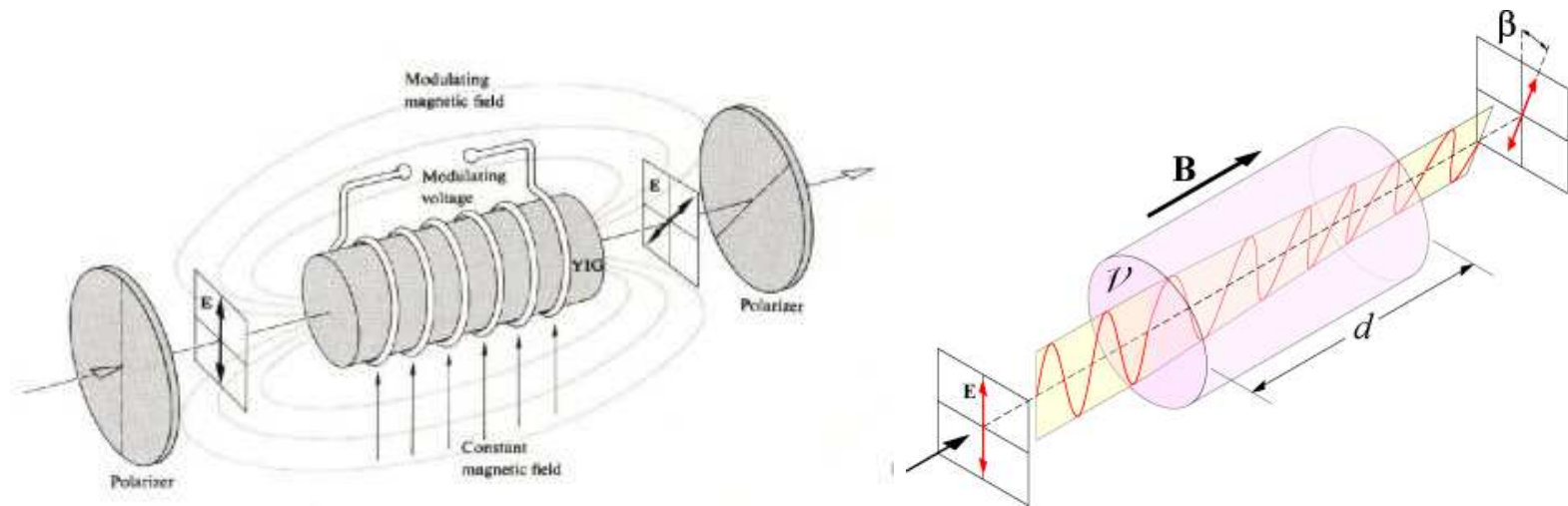
An arbitrary set of Stokes parameters can be represented as

$$\begin{pmatrix} I \\ Q \\ U \\ V \end{pmatrix} = \begin{pmatrix} I - \sqrt{Q^2 + U^2 + V^2} \\ 0 \\ 0 \\ 0 \end{pmatrix} + \begin{pmatrix} \sqrt{Q^2 + U^2 + V^2} \\ Q \\ U \\ V \end{pmatrix}$$

Thus, an arbitrary wave can be regarded as the independent superposition of a **completely polarized wave (first term)** and a **completely unpolarized wave (2nd term)**. A quasi-monochromatic is therefore said to be **partially polarized** with the **degree of polarization** defined as

$$p \equiv \frac{I_{\text{pol}}}{I} = \frac{\sqrt{Q^2 + U^2 + V^2}}{I}$$

Faraday Rotation: measuring B field



A magneto-optic effect, also known as the Faraday effect, in which the plane of polarization of an electromagnetic wave is rotated under the influence of a magnetic field parallel to the direction of propagation. It is named after the English physicist Michael Faraday (1791-1867), who first observed the effect in 1845.

Faraday rotation is displayed by radio waves as they travel through the interstellar medium. [Observing it in the radiation from pulsars is among the most important ways of studying the galactic magnetic field.](#)

Faraday Rotation: measuring B field

Assume that the interstellar gas is a tenuous plasma (dielectric medium with bound charges) with free electrons and ions and is magnetized. Only electrons need to be considered, since the motion of the ions is at least 3 orders of magnitude less. Also, external $\mathbf{B}$ field is much larger than the field strengths of the propagating wave, then eq. of motion of e in c.g.s unit:

$$m\dot{\mathbf{v}} = m\ddot{\mathbf{r}} = -e\left(\mathbf{E} + \frac{\mathbf{v}}{c} \times \mathbf{B}\right)$$

Assuming $\mathbf{B}$ is oriented in the z direction, i.e., the direction of the wave, then

$$\begin{aligned}\ddot{r}_x + \frac{eB}{mc}\dot{r}_y &= -\frac{e}{m}E_x \\ \ddot{r}_y - \frac{eB}{mc}\dot{r}_x &= -\frac{e}{m}E_y\end{aligned}$$

The input wave can be decomposed in R -handed and L -handed elliptically polarized as

$$E_{\pm} = E_x \pm iE_y$$

then we have

$$\begin{aligned}r_{\pm} &= r_x \pm ir_y \\ \ddot{r}_{\pm} \mp i\frac{eB}{mc}\dot{r}_{\pm} &= -\frac{e}{m}E_{\pm}\end{aligned}$$

Faraday Rotation

If $|E_x| = |E_y| = A$, i.e., **circularly polarized waves**, we have a more compact form

$$E_{\pm} = A e^{i(k_{\pm} z - \omega t)}$$

with A being a real number. Then we have a soln for r of the form

$$r_{\pm} = r_0 e^{i(k_{\pm} z - \omega t)}$$

Thus,

$$r_{\pm} = \frac{-\frac{e}{m}}{-\omega^2 \mp \frac{eB}{mc}\omega} E_{\pm}$$

and

$$\dot{r}_{\pm} = \frac{i\frac{e}{m}}{-\omega^2 \mp \frac{eB}{mc}\omega} \omega E_{\pm}$$

and the conductivity is purely imaginary,

$$\sigma_{\pm} = \frac{|\mathbf{J}|}{|\mathbf{E}|} = \frac{-Ne\dot{r}_{\pm}}{E_{\pm}} = i \frac{Ne^2}{m(\omega \pm \frac{eB}{mc})}$$

Faraday Rotation: Phase difference

$$\sigma_{\pm} = i \frac{Ne^2}{m(\omega \pm \omega_c)}$$

with the cyclotron or gyration frequency

$$\omega_c = \frac{eB}{mc}$$

For $\omega = \omega_c$, the freq. of the wave is in resonance with the gyration freq. of the electrons in the B field. Then $|\sigma_-| \rightarrow \infty$ and the E_- wave cannot propagate.

Remember the dispersion relation:

$$k^2 = \frac{\mu\epsilon\omega^2}{c^2} \left(1 + i \frac{4\pi\sigma}{\omega\epsilon}\right)$$

For a thin medium with $\epsilon \approx 1$ and $\mu \approx 1$, we have

$$k_{\pm}^2 = \frac{\omega^2}{c^2} \left(1 - \frac{\omega_p^2}{\omega(\omega \pm \omega_c)}\right)$$

with the plasma frequency

$$\omega_p^2 = \frac{4\pi Ne^2}{m}$$

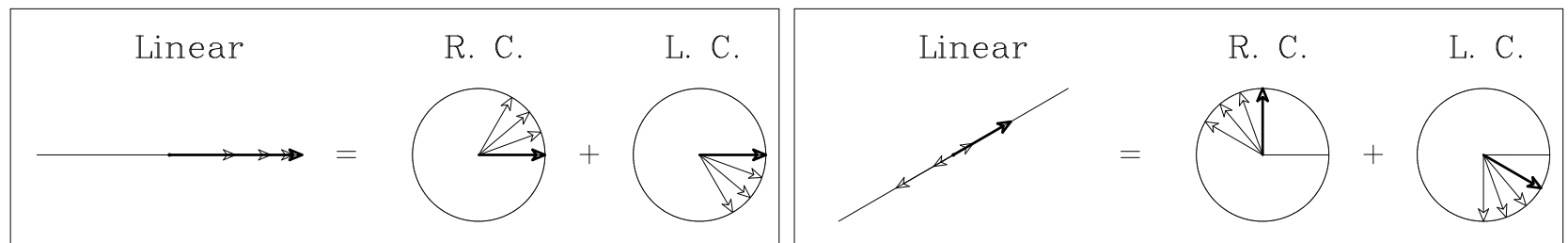
Faraday Rotation

The [index of refraction](#), which is given by $n = \frac{ck}{\omega}$, is

$$n_{\pm}^2 = 1 - \frac{\omega_p^2}{\omega(\omega \pm \omega_c)}$$

Thus, the two modes E_+ and E_- have slightly different phase propagation velocities $v_{\pm} = c/n_{\pm}$, with $v_+ > v_-$.

An elliptical polarized wave can be decomposed into two circularly polarized waves. [If the two circularly polarized waves have same \$E\$ field strengths, then the elliptical polarized wave becomes linearly polarized.](#)



Therefore, a plane (linear) polarized wave, which is a linear superposition of a right-handed and left-handed polarized wave, will not keep a constant plane of polarization, but this plane will [rotate](#) as it propagates ==> [Faraday rotation](#).

Faraday Rotation

Suppose we have a linearly polarized wave that decomposes into two circularly polarized components, then the two polarized wave components have a relative phase different $2\Delta\psi$ after a propagation length Δz :

$$2\Delta\psi = (k_+ - k_-)\Delta z$$

and the polarization angle of the linearly polarized wave will change by $\Delta\psi$.

From before,

$$k_{\pm}^2 = \frac{\omega^2}{c^2} \left(1 - \frac{\omega_p^2}{\omega(\omega \pm \omega_c)} \right)$$

If $\omega \gg \omega_p$ and $\omega \gg \omega_c$, then

$$k_{\pm} \approx \frac{\omega}{c} \left[1 - \frac{\omega_p^2}{2\omega^2} \left(1 \mp \frac{\omega_c}{\omega} \right) \right]$$

and thus

$$k_+ - k_- \approx \frac{\omega_p^2 \omega_c}{c\omega^2} = \frac{4\pi N e^3 B}{m^2 c^2 \omega^2}$$

$$\Delta\psi \approx \frac{2\pi N e^3 B}{m^2 c^2 \omega^2} \Delta z$$

Faraday Rotation

From before,

$$\Delta\psi \approx \frac{2\pi N e^3 B}{m^2 c^2 \omega^2} \Delta z$$

The total rotation of the polarization direction is

$$\Delta\psi = \frac{e^3}{2\pi m^2 c^2} \frac{1}{\nu^2} \int_0^L B_{||}(z) N(z) dz$$

with $\omega = 2\pi\nu$ and $B_{||}$ the B field along the line of sight. Explicitly, in λ ,

$$\frac{\Delta\psi}{\text{rad}} = 8.1 \times 10^5 \left(\frac{\lambda}{\text{m}}\right)^2 \int_0^{(L/\text{pc})} \left(\frac{B_{||}}{\text{Gauss}}\right) \left(\frac{N}{\text{cm}^{-3}}\right) d\left(\frac{z}{\text{pc}}\right)$$

Define a rotation measure (RM) as

$$\frac{\text{RM}}{\text{rad m}^{-2}} \equiv 8.1 \times 10^5 \int_0^{(L/\text{pc})} \left(\frac{B_{||}}{\text{Gauss}}\right) \left(\frac{N}{\text{cm}^{-3}}\right) d\left(\frac{z}{\text{pc}}\right)$$

we have

$$\frac{\Delta\psi}{\text{rad}} = \text{RM} \left(\frac{\lambda}{\text{m}}\right)^2$$

thus, $\Delta\psi \propto \lambda^2$. So measuring $\Delta\psi$ at two different wavelengths, we can derive RM.

Rotation Measure

The differential form is

$$\frac{d\psi}{d\lambda} = 2\text{RM}\lambda$$

Here positive RM indicates $B_{||}$ points towards us. For pulsar, we combine the values of RM and DM ($= \int N dz$). The resulting ratios give the

$$\frac{\bar{B}_{||}}{\text{Gauss}} = 1.23 \times 10^{-6} \frac{\text{RM}}{\text{DM}}$$

The galactic magnetic field $0.3\text{--}3 \mu\text{Gauss}$. The direction of $\bar{B}_{||}$ is away from the sun for $l = 0^\circ - 180^\circ$ and towards the sun for $l = 180^\circ - 360^\circ$ because of galactic rotation.